

ROBOTICS

MEEC

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Group 28

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Introduction

The objective of this second laboratory assignment is to implement a human augmentation device, that can be attached to a human body to assist with a 1 DOF (Degrees Of Freedom) linear motion. The specific application developed is a “button-pushing” assistant designed for PWD (Person/People With Disabilities), to maybe help push the button of an elevator or a light switch.

We opted to develop a system that allows precise control of a slider.

NOTE: For this assignment, all the software produced was written in *C++* and uploaded to the *Arduino Uno* using *PlatformIO*.

Task 1

The sensor used in this assignment is the *Grove - IMU 9DOF (lcm20600+AK09918)* IMU (Inertial Measurement Unit) [2]. It is composed of a 3-axis gyroscope, a 3-axis accelerometer and 3-axis electronic. The sensor is connected to the *Arduino Uno* board and communicates with it via the *I²C* protocol. To read and view the data from the sensor we used a web serial plotter [1] and the code *read_imu.ino* that was provided:

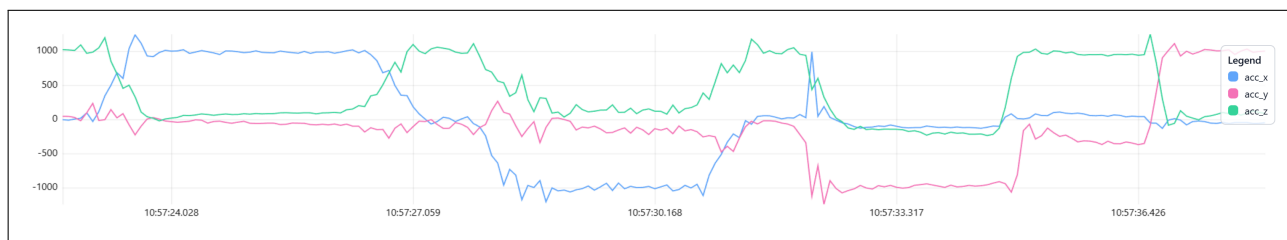


Figure 1: Visualization of the accelerations across the 3-axis.

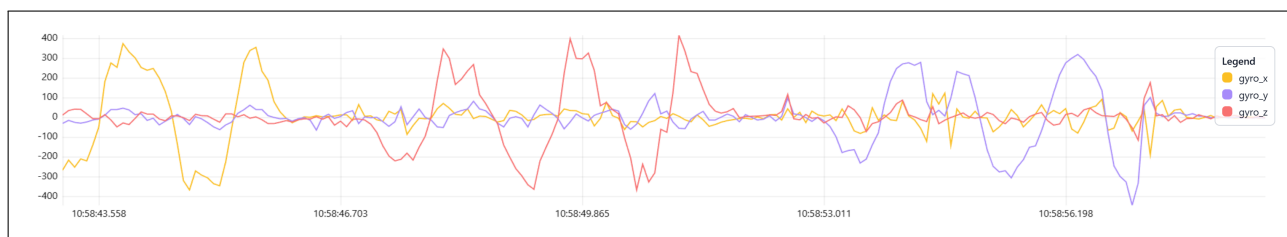


Figure 2: Visualization of the angular velocities around the 3-axis.

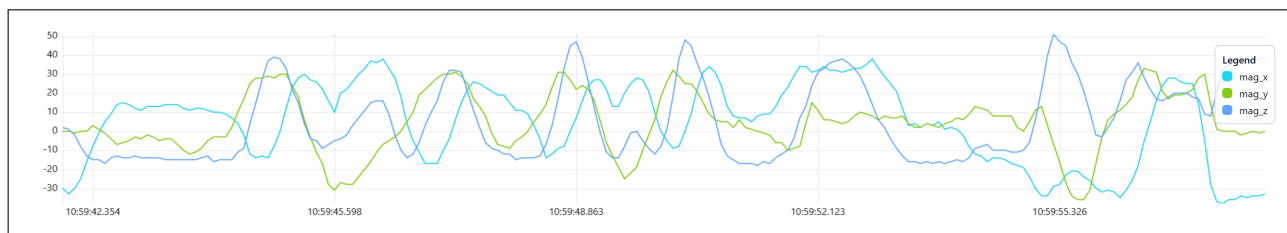


Figure 3: Visualization of the magnetic field across the 3-axis.

Task 2

Gesture definition To be able to move the slider by doing a gesture we first need to define which type of gesture we want our system to detect and then decide how we will move the slider according to the gesture. Since we want a precise movement of the slider, we decided that a solid approach would be control the velocity of the slider based on the pitch angle of the sensor board.

Motor control To control the slider we first need to control the motor, as they are mechanically linked. On other words, the velocity of the slider is directly related to the angular velocity of the motor, through these equations:

$$v_{belt} = r_{pulley} \cdot w_{motor} \quad \text{and} \quad v_{slider} = v_{belt}$$

Unlike DC motors, stepper motors work by dividing one full revolution into a fixed number of discrete angular steps. The stepper motor that we are using is the *NEMA 17* [4], which has 200 steps per revolution, resulting in a constant angular resolution of $\frac{360}{200} = 1.8$ degrees per step. Even though we cannot control directly the angular velocity of the motor, we can control the time interval between the steps (Δt), thus effectively changing its angular velocity (w_{motor}) and therefore changing the velocity of the slider (v_{slider}):

$$\text{smaller } \Delta t \Rightarrow \text{bigger } w_{motor} \Rightarrow \text{bigger } v_{slider}$$

Given that we want to use the pitch of the sensor board to control the angular velocity, the most natural way is to make the angular velocity directly proportional to the pitch, which translates to making the time interval between the steps inversely proportional to the pitch (θ):

$$\Delta t = \frac{C}{\theta}$$

C is just a scalar constant that was used to make a good relationship between Δt and θ , we empirically concluded that 50000 was appropriate.

Pitch estimation The last step is just to estimate the pitch angle with the measurements of the IMU. One approach is to just use the data from the accelerometer [3]. When the board pitches and rolls, the gravity vector will no longer just have a component on the z axis but will instead have a component on all three axis. This allows to have to estimate the pitch angle based on the value of the components a_x , a_y and a_z , that are the direct readings of the accelerometer on the IMU:

$$\theta = \arctan \left(\frac{a_x}{\sqrt{a_y^2 + a_z^2}} \right)$$

It is worth noting that this formula assumes that the IMU is stationary, meaning it only measures acceleration due to gravity. Even though there might be some lateral and vertical accelerations produced by the user, these should be subtle, meaning it won't affect much the pitch estimation. From our several tests we concluded that the estimation for the pitch angle was very good.

Sampling Frequency Coordinating the stepper motor introduces a challenge, it's speed is affected by the time it takes to run our loop function. For each iteration, the motor pin is set to high and back to low, with a certain delay. If there are other time consuming steps in our code, they will introduce an extra delay, slowing down the frequency of the motor steps. For this reason, we can not sample IMU data for every iteration. We set a sampling frequency of 50 Hz and verified that it did not have any noticeable affect on the responsiveness of the system and ensured accurate speed control.

Deadzone To improve the user experience, we set a dead-zone of $[-10; 10]$ degrees. If the calculated pitch angle is within this interval, the motor won't rotate. This guarantees that the slider does not move with slight unintentional movements.

Safety Limits The system setup includes a push button at each end of the pulley. Trying to move the slider past these limits can damage the equipment, so we decided to implement a function that pulls the slider back when it activates a button.

Task 3

In this task, the objective is to make the slider move between the extreme switches and stop at the middle. To do this we simply move the slider all the way to one side (we choose the left one) then move in the opposite direction and count the number of steps that were taken until we reach the other end. Since we know how much steps the motor took to traverse from side to side, we can simply divide that value by 2 and move that many steps in the original direction to reach the middle. Here is the demonstration of the whole process:

<https://drive.google.com/file/d/1EDfdd0tvxSyPfXPEQAMd6m99kneWJF-C/view?usp=sharing>

Task 4

For the last task, we filmed a video demonstrating how our systems behaves based on on the pitch angle of the IMU board:

<https://drive.google.com/file/d/1fWGJNN3JpK6hZQPbsAn6FpMCcF4GJXSs/view?usp=sharing>

Role of each author

Beatriz Coelho: Stepper motor control
Daniel Burke: Sensor data processing
Diogo Sousa: Software
Miguel Fialho: Kinematics

References

- [1] atomic14. Web serial plotter.
- [2] Farnell. *Grove - IMU 9 DOF Datasheet*.

- [3] Inertial Labs. Precision pitch and roll measurement.
- [4] PCB Linear. *NEMA 17 Stepper Motor Datasheet*.